# Report on "Ancilla-assisted protection of information: application to atom-cavity systems"

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# 1 Problem Introduction

In modern quantum-enabled technologies, we need well-controlled systems like atomic devices and solid-state setups to store and process quantum information. However, one of the biggest fundamental problems in quantum mechanics is that we cannot fully isolate a quantum system from its environment. This inevitable interaction causes environmental noise, which leads to decoherence. Decoherence actively destroys the quantum aspects of the state, such as quantum coherence and entanglement, and introduces uncontrollable errors into the system while it undergoes quantum operations [1].

While many techniques exist to mitigate this environmental noise, they come with severe limitations. For instance, dynamical decoupling is a popular open-loop quantum control technique that uses periodic instantaneous control pulses, but it completely fails when applied to non-Markovian processes like amplitude damping [1]. Furthermore, standard quantum error correction requires a massive overhead of extra auxiliary qubits and complicated measurement feedback loops [2]. Other techniques like the quantum Zeno effect or weak measurement reversals suffer from low success probabilities due to measurement failure rates. Therefore, it is highly necessary to find a practical and simpler way to protect a quantum state indefinitely. This report discusses a novel protocol that makes an arbitrary noisy environment effectively noise-free and transparent to the system by using an ancilla, specifically photons, to protect the information stored in an atomic system.

# 2 Methodology and Experimental Procedure

## 2.1 Atom-Cavity System Setup

The experimental methodology focuses on a scheme where a single three-level atom is trapped inside an optical cavity. The purpose of this cavity setup is to protect the atomic state for an indefinite amount of time. The main system (S) is the three-level atom, where the computational basis is formed by the low-energy degenerate ground states $|\pm 1\rangle$, and there is one excited state $|0\rangle$. The ancilla system consists of two single photons, which we will label as photon A and photon B.

To initiate the physical procedure, the photons are directed into the optical cavity using circulators, where they interact with the trapped atom. As shown in Figure 1, the interaction heavily relies on the polarization of the incoming photons and the conservation of angular momentum. Right-circular polarized light interacts specifically with the

$|+1\rangle \leftrightarrow |0\rangle$ transition, while left-circular polarized light interacts exclusively with the $|-1\rangle \leftrightarrow |0\rangle$ transition [3].

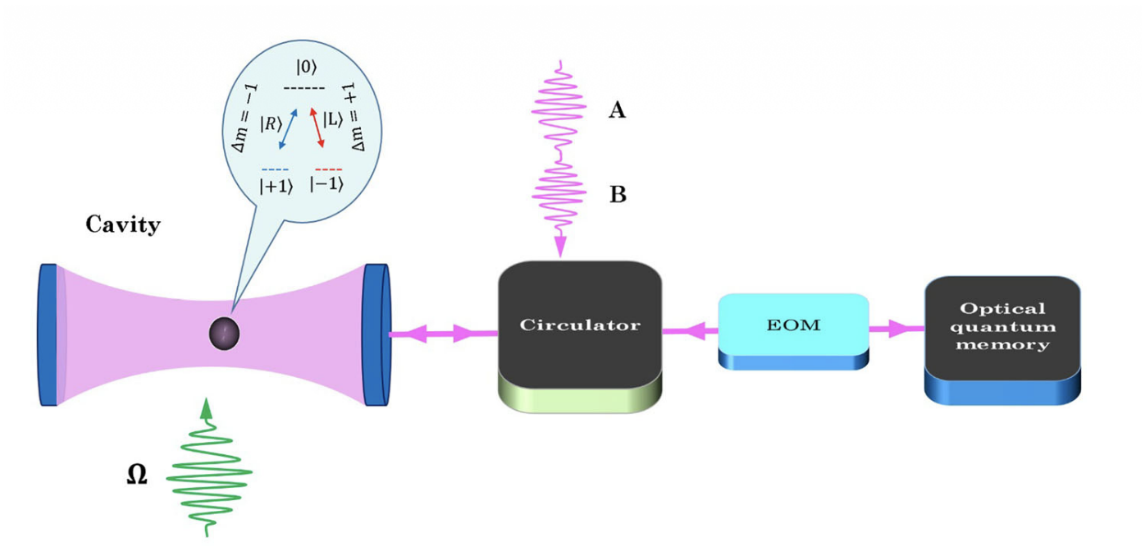


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Figure 1: Experimental setup for the noise mitigation protocol. A three-level atom is trapped inside an optical cavity. Ancillary single-photons A and B interact with the atom sequentially via circular polarization transitions to perform controlled unitary operations. Circulators direct the photons, while an EOM and optical quantum memory assist in the subsequent decoding and storage phases.

## 2.2 Unitary Encoding Implementation

The core methodology requires operating the cavity in the strong coupling regime, also known as the Purcell regime. In the steady-state limit and at resonance, the relationship between the input light mode $a_{in}$ and the output light mode $a_{out}$ is defined mathematically as:

$$a_{out} = \frac{-\kappa\gamma + 4g^2}{\kappa\gamma + 4g^2} a_{in} \tag{1}$$

where $\kappa$ is the decay rate of the cavity, $g$ characterizes the coupling strength between the cavity and the atom, and $\gamma$ is the atomic decay rate. The protocol requires $\kappa \gg g \gg \gamma$.

First, photon B is sent into the cavity. Because the transition $|-1\rangle \leftrightarrow |0\rangle$ is decoupled for right-circular polarization, an atom in state $|-1\rangle$ will reflect the photon with a $\pi$-phase shift. This physical interaction results in a controlled-Z operation. Next, a Hadamard gate is applied to the atom using classical laser pulses via the optimized Stimulated Raman Adiabatic Passage (STIRAP) technique [4]. Finally, photon A interacts with the cavity in the exact same manner, applying a controlled-X operation. Together, these physical steps form the encoding unitary operation $U_{ABS}$. The photons are subsequently routed to an optical quantum memory for safe storage, while the atom remains exposed to the harsh environmental noise.

# 3 Results: Step-by-Step State Evolution

## 3.1 Initial Preparation and Pre-processing

The primary result of this protocol is the successful, deterministic protection of the atomic quantum state. We demonstrate this result by tracking the exact mathematical evolution of the density matrices throughout the quantum circuit, which is visually represented in Figure 2.

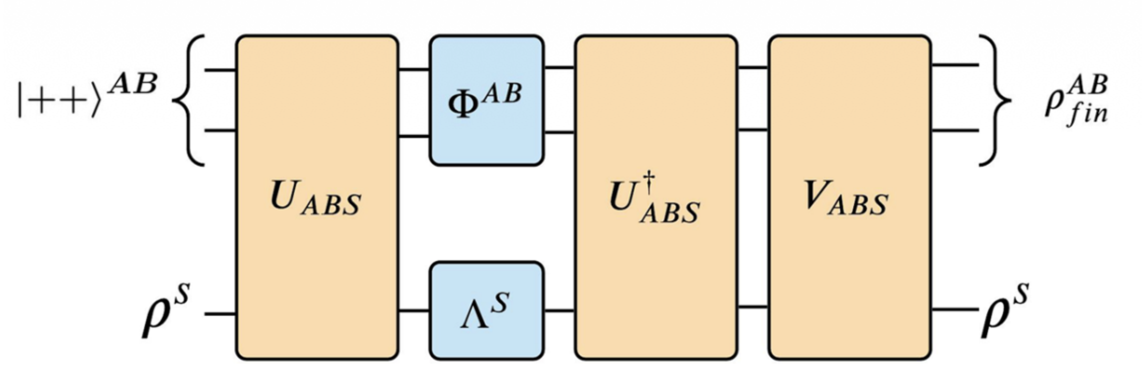


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Figure 2: Quantum circuit diagram for the noise mitigation protocol. The system S couples with ancilla AB via unitary $U_{ABS}$ before undergoing arbitrary environmental noise $\Lambda^S$. The ancilla is exposed to restricted noise $\Phi^{AB}$. Finally, the inverse operations and correction unitary $V_{ABS}$ neutralize the noise effect, completely restoring the system state.

In the very first step, the initial setup is prepared. The ancilla photons AB are initialized in the superposition state $|{+}{+}\rangle^{AB}$, where $|+\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$. The atom S is prepared in an arbitrary initial pure state $\rho^S$. The resultant state of the entire composite system at this starting stage is simply the tensor product of the two independent systems. The resultant state is written mathematically as:

$$\rho_1^{ABS} = |{+}{+}\rangle \langle{+}{+}|^{AB} \otimes \rho^S \tag{2}$$

## 3.2 Information Transfer and Decoherence-Free Subspace

Next, the system undergoes the encoding process. The composite system is evolved using the global unitary operation $U_{ABS}$, which we physically implemented in the methodology section using cavity reflections. This operation couples the atom and the photons before any environmental noise can affect the atom. The unitary operator is mathematically defined by the following expression:

$$U_{ABS} = |00\rangle \langle 00|^{AB} \otimes I^S + |01\rangle \langle 01|^{AB} \otimes \sigma_z^S + |10\rangle \langle 10|^{AB} \otimes \sigma_x^S - i\,|11\rangle \langle 11|^{AB} \otimes \sigma_y^S \tag{3}$$

The resultant state of the global composite system after this unitary evolution is computed as:

$$\rho_2^{ABS} = U_{ABS}\rho_1^{ABS}U_{ABS}^{\dagger} \tag{4}$$

If we analyze the resultant state of the photons at this point by taking the partial trace over the atomic system S, the reduced density matrix of the ancilla AB becomes:

$$\rho_{AB} = \frac{1}{4}\left[(I \otimes I) + r_z(I \otimes \sigma_x) - r_y(\sigma_x \otimes \sigma_y) + r_x(\sigma_x \otimes \sigma_z)\right] \tag{5}$$

This represents a highly significant theoretical result. The resultant system demonstrates that all the initial information about the atomic state, specifically the Bloch vector components $r_x, r_y$, and $r_z$, has been completely extracted from the atom and transferred into the bipartite correlations of the two photons. Simultaneously, the resultant state of the atom S is now completely maximally mixed, meaning it effectively holds zero coherent information locally.

## 3.3 Environmental Interaction

Following this successful encoding, the atom S is exposed to its surroundings and undergoes an arbitrary, completely unknown noisy quantum channel $\Lambda^S$. This channel can be expressed through Kraus operators $F_m^S$. Simultaneously, the ancilla photons might also undergo a specific class of environmental noise while stored in the optical quantum memory, denoted by the channel $\Phi^{AB}$. The resultant state of the composite system after experiencing the combined environmental noise is:

$$\rho_3^{ABS} = \Phi^{AB} \otimes \Lambda^S(\rho_2^{ABS}) \tag{6}$$

A major finding of this theoretical derivation relates to the concept of the decoherence-free subspace. Even though the photons experience noise modeled by $\Phi^{AB}$, their resultant state $\rho^{AB}$ does not actually change. The mathematical property of this specific evolution dictates that $\rho^{AB} = \Phi^{AB}(\rho^{AB})$. This happens because the information was encoded into a specific subspace that satisfies the condition where the commutator of the state and the noise Kraus operators is zero [5]. Therefore, the resultant state of the stored quantum information is perfectly preserved in the photonic correlations even while the atom is being actively destroyed by arbitrary noise.

## 3.4 Decoding and Post-processing

To retrieve the information, the decoding step must be systematically applied. The combined system is evolved using the inverse unitary operation $U_{ABS}^\dagger$. The resultant state of the system after this specific decoding step is calculated as:

$$\rho_4^{ABS} = U_{ABS}^\dagger \rho_3^{ABS} U_{ABS} \tag{7}$$

This resultant step successfully pulls the quantum information back from the photonic correlations and directs it towards the atom. However, the noise effect on the atom has not yet been fully neutralized.

Finally, a post-processing correction operation $V_{ABS}$ is applied to the system. This operation is built using Hadamard gates applied to the polarization states of the photons and the atomic states. The final resultant state of the total composite system after all operations are completed is given by:

$$\rho_{fin}^{ABS} = V_{ABS} \rho_4^{ABS} V_{ABS}^\dagger \tag{8}$$

The ultimate resultant state of the atom S, when we mathematically trace out the photons from the final density matrix, is exactly $\rho^S$. This conclusively proves that the arbitrary initial state of the atom is fully recovered without any loss of fidelity. The devastating effect of the arbitrary noisy channel $\Lambda^S$ on the atom has been completely transformed into an identity operation $I^S$. The protocol succeeds entirely in making the chaotic environment completely transparent to the protected atom.

# 4 Analysis, Discussion, and Conclusion

## 4.1 Comparison with Traditional Error Correction

The theoretical and experimental results of this protocol show a highly effective method for protecting quantum information that heavily differs from standard approaches. In traditional Quantum Error Correction (QEC) schemes, a significant number of extra auxiliary qubits are required to encode a single logical qubit. Furthermore, the system must undergo frequent intermediate syndrome measurements to detect and actively fix errors. These continuous measurements and feedback loops make the experimental process highly complicated and prone to further noise induction. In stark contrast, this proposed protocol requires only two ancillary photons and operates entirely without intermediate measurements. The entire recovery process is achieved using only controlled unitary operations, keeping the evolution completely coherent from start to finish [1].

Another major advantage analyzed in this study is the deliberate separation of noise models. The protocol allows the atom to undergo literally any arbitrary noise channel, regardless of its mathematical complexity, as long as the ancilla photons undergo a restricted class of noise. This fits perfectly with realistic physical laboratory setups because atoms and photons naturally experience very different types of noise environments even when they are physically located in the exact same macroscopic room.

## 4.2 Impact of Photon Loss

However, we must strictly analyze the physical limitations and potential failure modes of the experimental setup. A primary concern with using flying qubits like photons as an ancilla is the probability of complete photon loss. Current state-of-the-art atom-cavity experiments show that photon loss can be restricted to about 5% [6]. We must consider what happens if a photon is lost during different stages of the protocol.

If photon A is lost before step 2 takes place, the effect of environmental noise on the atom cannot be eliminated, and the atom undergoes an uncontrolled evolution. If a photon is lost with a probability $p$, the overall evolution of the atom becomes a mixed channel $\mathcal{E}^S = (1-p)I^S + p\Phi_2^S$. The error incurred in the noise mitigation due to this photon loss can be quantified using the diamond norm metric. The error is strictly bounded from above by the probability of loss:

$$||I^S - \mathcal{E}^S||_\diamond = p||I^S - \Phi_2^S||_\diamond \leq p \tag{9}$$

This bound holds true whether photon A is lost in step 2 or step 4, and is symmetrically true for photon B [7]. Furthermore, if both photons A and B are lost with independent probabilities $p$ and $q$ respectively, the overall error in the noise mitigation protocol is upper bounded by the product $pq$, irrespective of the specific protocol steps in which the photons disappear.

### 4.3 Fidelity and Timing Constraints

In addition to photon loss, the controlled operations between the atom and the optical cavity are not perfectly ideal. For example, a standard CNOT operation in such systems currently has an 86% fidelity. This reduction in fidelity primarily arises from frequency fluctuations and scattering losses associated with the cavity mirrors, rather than the cavity decay rates themselves [3].

The time scales required for the experiment also play a crucial role in the protocol's viability. Single-qubit atomic rotations performed using the STIRAP technique are extremely fast, taking only about 100 ns. However, the total time required for the controlled operations like $U_{ABS}$ takes around 5 $\mu$s because it fundamentally depends on the input pulse width of the photon. By significantly boosting the cavity coupling strength $g$, it is physically possible to speed up the execution of these controlled operations, bringing the total interaction time down to a few hundred nanoseconds, thus minimizing the window where uncontrollable decoherence can heavily act on the system.

### 4.4 Conclusion

In conclusion, this report comprehensively demonstrates that an arbitrary noisy environment can be made completely transparent to an atomic quantum system. By transferring the atom's state information into the specifically tailored decoherence-free subspace of two ancillary photons, the quantum information can survive severe, unknown environmental interactions. Once the environment has acted upon the system, the information is retrieved back to the atom without loss of coherence or entanglement. Despite current physical experimental limitations regarding gate fidelities and inevitable photon loss via scattering, this protocol provides a powerful, robust, and measurement-free alternative to traditional quantum error correction architectures.

## References

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